

Formal Verification Supplement: Finite Aggregation Accuracy

1 Verified accuracy of finite good aggregations

For $u, v \in \mathbb{R}^r$, $r \geq 2$, define

$$f(u, v) = (\|u\|^2 - 1, \|v\|^2 - 1, 1/2 - u^\top v), \quad S = \{x : f_i(x) < 0 \text{ for all } i\}, \quad C = \text{cl}(\text{conv } S).$$

The preceding verified classification identifies good multipliers exactly as nonzero $\lambda \geq 0$ with $\lambda_3^2 \leq 4\lambda_1\lambda_2$. Write $f_\lambda = \sum_i \lambda_i f_i$. For a finite set F of good multipliers, put

$$P_F = \{x : f_\lambda(x) \leq 0 \text{ for every } \lambda \in F\}, \quad e_N = \inf_{|F| \leq N} d_H(C, P_F).$$

Here d_H is the extended Hausdorff distance in the Euclidean norm on all $2r$ original coordinates. It is infinite for an unbounded relaxation. The infimum ranges over arbitrary good multipliers, including interior ones and arbitrary positive rescalings; no minimizing family is assumed.

The Lean package proves, for every $N \geq 2$,

$$\frac{\sqrt{2}(\log 2)^2}{1600N^2} \leq e_N \leq \frac{5\sqrt{2}\pi^2}{16(N-1)^2}.$$

The constants are independent of r . A linear homeomorphism into an explicit Euclidean space proves agreement with the original variables, their quadratic residuals, and their actual closed convex hull. In particular the default maximum norm on a function space is not used in these error bounds.

A uniform upper bound. The angle multipliers

$$\lambda(\theta) = (\cos^2 \theta, \sin^2 \theta, 2 \sin \theta \cos \theta), \quad 0 \leq \theta \leq \pi/2,$$

are good, and all their weak inequalities describe C . The formal proof includes the coordinate endpoints and does not require the associated two-by-two slack matrix to be PSD: only its nonnegative directions are tested.

For a sample set with covering radius ρ , the angular quadratic $h_x(\theta) = -f_{\lambda(\theta)}(x)$ is bounded below by $-5\rho^2$ throughout the interval whenever the sampled cuts and both coordinate bounds hold. The proof minimizes this continuous function. At an interior minimum, its derivative is zero; an exact rotation identity and $|\sin t| \leq |t|$ give the bound. Endpoint minima are already nonnegative. This is an elementary alternative to the paper's second-derivative Taylor estimate.

With $a = 5\rho^2$, contraction by $s = (1 + 2a)^{-1/2}$ repairs every point of the relaxation into C . The Euclidean displacement is at most $\sqrt{2}a$. The N equally spaced angles have covering radius $\pi/(4(N-1))$, giving the displayed upper constant. The estimate concerns the entire relaxation, not only selected witnesses.

A finite-grid lower bound. The verified lower proof uses $N + 1$ parameters $\tau_j = 1 + j/N$, $j = 0, \dots, N$, and the rational perturbation $\eta = 1/(400N^2)$. It realizes the Gram matrices

$$\begin{pmatrix} 1 - 1/(10\tau_j) & 2/5 - \eta \\ 2/5 - \eta & 1 - \tau_j/10 \end{pmatrix}$$

by actual vectors in every required dimension. If one good multiplier excludes witnesses at two parameters $\tau, \sigma \in [1, 2]$, the cone inequality gives

$$(\tau - \sigma)^2 < 16((1 + 10\eta)^2 - 1) < 1/N^2.$$

Thus a cut excludes at most one grid point, and at most N cuts leave one witness admitted. The argument includes zero-coordinate and interior multipliers and imposes no bound on their ratios.

At an admitted witness, the good multiplier $(\tau, 1/\tau, 2)$ has value 2η . Its quadratic has Lipschitz constant at most 10 on the product of the two Euclidean unit balls, where both the witness and C lie. Every hull point therefore has distance at least $\eta/5$ from the witness. This proves the stronger lower estimate

$$d_H(C, P_F) \geq \frac{1}{2000N^2} \geq \frac{\sqrt{2}(\log 2)^2}{1600N^2}.$$

The final comparison is itself proved in Lean. The main paper improves the first lower constant to $\sqrt{2}/2000$ using a sharper Lipschitz estimate; that improvement is not part of this formal package. This proves the lower bound by a finite pigeonhole argument, without requiring an interval-covering theorem.

Exact rational coefficients. For $m \geq 1$, use

$$(m^2, j^2, 2mj), \quad (j^2, m^2, 2mj), \quad 0 \leq j \leq m.$$

The common middle ray is counted once, leaving exactly $2m + 1$ distinct cuts. They are good and include both coordinate rays. Their angles have covering radius at most $1/(2m)$, proved using the Lipschitz bound for arctangent. Positive rescaling connects these exact integer coefficients to the angle-family theorem, yielding error at most $5\sqrt{2}/(4m^2)$.

For $N \geq 3$, choosing $m = \lfloor (N - 1)/2 \rfloor$ meets the cut budget. Each coefficient is a nonnegative integer at most $2m^2$. With the binary size convention assigning zero size to zero, the package bounds each size by $2 \text{size}(N) + 1$, and hence by $2 \lfloor \log_2 N \rfloor + 3$. It also proves a uniform constant times N^{-2} error bound for these rational families. These statements concern exact representation; they do not analyze numerical rounding or solver conditioning.

Rates, tolerances, and scope. The extended-real family bounds imply finiteness of e_N for $N \geq 2$ before it is converted to a real number. The package then proves the asymptotic $\Theta(N^{-2})$ statement. Writing $c = \sqrt{2}(\log 2)^2/1600$ and $B = 5\sqrt{2}\pi^2/16$, any family of at most $N \geq 1$ cuts with error at most $\varepsilon > 0$ must satisfy $N \geq \sqrt{c/\varepsilon}$. Conversely the explicit angle family with at most $\lceil \sqrt{B/\varepsilon} \rceil + 1$ cuts suffices; its size is less than $\sqrt{B/\varepsilon} + 2$. Sufficiency constructs a family and does not infer attainment from an infimum estimate.

The frozen claims and declaration coverage are in `supplement/lean/audits/31-aggregation-accuracy/`. The support-function interpretation and single-objective exact-aggregation proposition in the main paper are outside this package. These uniform approximation bounds give no solver runtime, iteration, or branch-and-bound lower bound, and make no literature-priority claim.